

DECODED EXPLAINED

27th August, 2026

Why India's R&D System Needs a Map of Where Funds Really Go

Why India's R&D system needs a map of where funds really go

India does not have a system that can tell funders, researchers, or the public details like who is being funded, by whom, for what, and whether that funding has already been given elsewhere; while the NITI Aayog has proposed a unified streamlining system, it still fails to address a key problem

Mousmita Koley

A recent report by the NITI Aayog, about 'Ease of Doing R&D in India' revealed some uncomfortable truths about the research and development ecosystem in the country. Together with a companion survey with inputs from more than 400 institutional leaders and 850 scientists, the report said that close to 80% of funding under the Anusandhan National Research Foundation (ANRF) – the flagship research funding body – is concentrated in the ITs. This is despite the ANRF's mandate to support a wider base of universities and research bodies than the traditional funding model has managed.

There are also concerns that multiple Central agencies are possibly funding similar research areas, resulting in overlap and what the report calls "inefficient" use of public money. These findings point to a common underlying gap: India does not have a system that can tell funders, researchers, or the public who is being funded, by whom, for what, and whether that funding has already been given elsewhere.

The NITI Aayog's response to this gap is the Unified Project Management System (UPMS) – a system to streamline the planning, funding, monitoring, and evaluation of public R&D projects across Ministries and Departments.

This proposed solution sounds good; however, it does not directly address an important problem.

PIDs and metadata

The technical term for what is missing is PIDs and metadata infrastructure. In plain language, it means every research grant in India will need a permanent, unique, machine-readable identifier called a persistent digital identifier (PID). This is like how a PAN number identifies a taxpayer or an IMEI number identifies a mobile phone.

Each PID should then be attached to a standard set of details called the metadata, which agency gave the money, to which institution, to which named researcher, what amount, over what period, and in which field. Finally, there is a linking system that will allow officials to track these details using the PIDs.

So every time a scientist publishes a paper and includes the PID in it, the system can track and evaluate the outcome of public funding. At least this is the idea.

Right now, a lot of this information exists within Indian funding agencies – but it is scattered across dozens of separate agency databases, in inconsistent formats, and often as free text rather than structured data. A researcher's name might appear one way in a Department of Science and Technology (DST) record and slightly differently in a Council for Scientific and Industrial Research (CSIR) database – and this difference can be enough for an automated system to not tell that they are the same person. An institution might be listed as "IISc, Bangalore" in one dataset and "Indian Institute of Science, Bengaluru" in another, leading to a similar problem.



Currently, information about the use of public funding for research lies scattered across dozens of separate agency databases, in inconsistent formats, and often as free text rather than structured data. Representative image. [TALHA HASAN/UNISPLASH](#)

Multiply such inconsistencies across thousands of grants a year and one can see that the result is an information that does not provide a reliable picture of how research funding is being used.

Other countries' solutions

This is not a new problem and India does not have to solve it from scratch. A global, non-profit infrastructure setup called the Grant Linking System already exists to do exactly this. It was built by the open digital infrastructure organisation Crossref. Its registry is called 'Research Organisation Registry' (ROR), built by a different organisation.

The system works through a small number of PIDs, each solving one piece of the puzzle. First, a funder ID identifies the funding agency in detail – the division, department, and even ministry if required. A similar ROR ID identifies the institution receiving the funding. Finally, an ORCID identifier – familiar to Indian researchers today since most scientific journals today ask for it – identifies the individual researcher, so that funding data can be aggregated per person.

Since 2020, Crossref has also extended its system to include a grant DOI: a permanent identifier for the grant itself, which can be linked to whatever comes out of it – papers, patents, data.

Taken together, these four identifiers plus the metadata allow a funder or an independent analyst to ask questions that are nearly impossible to answer in India today. For example, how many different agencies have funded near-identical proposals? Which fields are receiving disproportionate investments relative to the national priorities? Does a small number of institutions or researchers account for a large share of the total

funding? This is why many research funders have adopted this system, with more than 2 lakh grants registered on it worldwide. The scale of the underlying problem this system solves is also well-documented. One 2013 analysis of U.S. funders used automated text-matching software to scan several lakh federal grant applications and estimated that duplicate or substantially overlapping funding may have cost the U.S. nearly \$70 million. In a more recent analysis, in 2020, analysts checked nearly 20,000 competitive grants in Denmark and found that funds were getting concentrated among a small group of individual researchers and a narrow set of research topics.

This is similar to the pattern the NITI Aayog report flagged for Indian funding agencies – and painstakingly so.

Two ways forward

Policymakers considering how to act on NITI Aayog's findings face two broad options. One is to build an entirely new, India-specific system from the ground up as a sovereign national grant registry, developed and maintained independently. This has the appeal of full control: India will own the infrastructure outright and could tailor it precisely to its own agencies and federal structure, including State research schemes that a global system might not naturally prioritise. The second option is to plug into the existing global non-profit infrastructure, requiring India's funding agencies to have funded members of – and consistently deposit data with – the Crossref ecosystem that many research funders already use worldwide. This can be implemented faster since the technical standards and governance already exist,

and are maintained by an international community rather than by a single Indian agency operating alone. It would also make India's publicly funded research immediately visible and comparable within the same global dataset used to study research funding everywhere.

There is also a practical middle path: using the technical infrastructure of PID generators and funders' own governance structure. Specifically, we can learn from systems that have worked elsewhere, such as the U.K. 'Gateway to Research' portal and the European Union's Community Research and Development Information Service (CORDIS) and Open Access Infrastructure for Research in Europe (OpenAIR). They run their own government-owned portals as the single point of entry for funding agencies, and the portals generate identifiers fully compatible with the global Crossref/ROR/ORCID standards.

This gives national governments political ownership and the ability to require all domestic agencies to comply – while also ensuring that the resulting data speaks the same language as the rest of the world's research infrastructure. For India, this hybrid model would mean a single national portal – one that the NITI Aayog has already proposed, called UPMS, and which Central and State funding agencies must use. The UPMS system can internally mint Crossref-compatible grant DOIs and link every record to ROR and ORCID identifiers. Then, it will be much easier to check for duplicate records, fund concentration, and so on.

(Mousmita Koley is a senior research analyst at the Indian Institute of Science, Bengaluru. mousmitakoley@iisc.ac.in)

Why India's R&D System Needs a Map of Where Funds Really Go

- **The Problem**

- A NITI Aayog report on 'Ease of Doing R&D in India', backed by a survey of over 400 institutional leaders and 850 scientists, found close to 80% of funding under the Anusandhan National Research Foundation (ANRF) is concentrated in the IITs, despite its mandate to support a wider base of universities.
- There are also concerns that multiple Central agencies may be funding similar research areas, resulting in overlap and what the report calls 'inefficient' use of public money.

- **The Underlying Gap**

- These findings point to a common gap: India lacks a system that can tell funders, researchers, or the public who is being funded, by whom, for what, and whether that funding has already been given elsewhere.
- NITI Aayog's proposed response is the Unified Project Management System (UPMS), meant to streamline planning and monitoring of public R&D but this doesn't directly address a deeper missing piece: 'PIDs and metadata infrastructure'.

- **How Other Countries Solve This**

- A global non-profit infrastructure called the Grant Linking System, built by Crossref, already solves this using a funder ID, a Research Organisation Registry (ROR) ID, and an ORCID identifier with over 2 lakh grants registered worldwide.
- The scale is documented elsewhere too: a 2013 US analysis put duplicate federal grant costs near \$70 million.

How Can States Use Disaster Funds for Heatwaves?

How can States use disaster funds for heatwaves?

Heatwave funding expands, but States must strengthen planning and implementation to use it effectively

Shravan Prabhu

On August 4, the Ministry of Home Affairs told the Lok Sabha that heatwaves and lightning had been added to India's list of notified natural calamities, taking the total list to 14.

The change follows a recommendation by the Sixteenth Finance Commission (FC-XVI) and operational guidelines issued on June 30. Along with the 2024 amendment to the Disaster Mitigation Fund guidelines, the update means heatwaves are now eligible for the full State Disaster Risk Management Fund pool.

Disaster finance in India

Under Article 280 of the Constitution, the Finance Commission recommends arrangements for disaster management funds. The FC-XVI recommended ₹2.04 lakh crore for State disaster funds in the five years between 2026-27 and 2030-31,

nearly 28% more than the previous Commission's allocation.

Of the total fund, ₹1.6 lakh crore goes to the State Disaster Response Fund (SDRF) and the rest to the State Disaster Mitigation Fund (SDMF). The SDRF is meant for response, relief, and reconstruction immediately following a disaster; the SDMF is meant to fund interventions to reduce the risk of a climate event from turning into a disaster, in the longer term.

These funds are available to all States. The FC-XVI has determined the inter-State distribution using a formula comprising a disaster risk index, which takes into account the frequency and intensity of different hazards along with exposure and vulnerability and expenses incurred by States during the previous commission's tenure. The FC-XVI recommended ₹79,406 crore for national disaster funds over the same period and which States can seek when a disaster is severe enough to exceed their own resources, like Kerala

did following the Wayanad landslides in 2024.

Heat has thus far occupied an awkward position in the disaster finance system. A State could notify a heatwave as a 'local disaster' and use the SDRF – but only up to 10% of its annual allocation and after setting separate norms such as the amount of compensation for loss of lives or damage to crops. Other notified disasters such as floods and cyclones faced no such ceiling. The new notification removes this asymmetry.

India's exposure to extreme heat is growing. The author used 35 indicators across climate change, exposure, and vulnerability to estimate that more than 57% of Indian districts, home to roughly three-fourths of the Indian population, face high- to very-high heat risk. The number of hot days per year has increased. The incidence of very warm nights and relative humidity has also risen, reducing the human body's ability to recover after hot days.

What States can fund

The National Disaster Management Authority has asked all States, districts, and cities to develop heat action plans (HAPs), which map heat risk and vulnerability and identify short-, medium-, and long-term interventions. Thus far, 300 cities and districts across 23 heatwave-prone States have HAPs.

Implementation, however, has lagged because of limited technical capacity, competing priorities, and inadequate funding. A 2023 review of HAPs also found that 79% of the plans asked city departments to self-allocate funds.

The new Central notification provides States with two clearer financing options. Under the SDRF, they can provide relief and compensation for heat-related losses. And through the SDMF, they can finance longer-term risk reduction such as cooling shelters and early-warning systems. The respective State Disaster Management Authorities and State Emergency Operations Centre will be the nodal agencies responsible for both funds' corpus.

Next steps

First, heat planning must expand. Twelve States have already notified heatwaves locally but only some 300 cities and districts have HAPs, leaving around 4,800 urban local bodies and 800 districts without them. Heat-specific SDMF guidelines are still awaited, but existing guidelines require risk and vulnerability

assessments before any projects are proposed – precisely the groundwork HAPs are meant to provide.

Second, plans must become fundable projects. States need model mitigation proposals suited to local climate and geography. Many local bodies lack the technical capacity to prepare such proposals. Research institutions, NGOs, and philanthropy could provide technical support and risk capital to help convert plans into appraisable projects.

Third, health surveillance must improve. Relief payouts will now depend on accurate loss-and-damage data. The Health Ministry's current surveillance system covers over 51,000 reporting units and recorded 4,853 heatstroke cases and 20 confirmed heatstroke deaths between March 1 and July 26. But it misses the wider burden from heart, lung, and kidney conditions worsened by heat.

Fourth, States should link heat solutions to other government development schemes by identifying convergence, and use fast, pre-agreed tools like parametric insurance, which pays out automatically once a set temperature threshold is crossed. Similar tools already operate for other hazards; Nagaland, for example, has rainfall insurance that pays out when rainfall levels cross a set threshold.

(Shravan Prabhu is Programme Lead at Council on Energy, Environment and Water (CEEW) and Coordinator of the GHHN South Asia Heat & Health Hub)

How Can States Use Disaster Funds for Heatwaves?

- Under Article 280 of the Constitution, the Finance Commission recommends arrangements for disaster management funds.
- The FC-XVI recommended ₹2.04 lakh crore for State disaster funds over the five years from 2026-27 to 2030-31 nearly 28% higher than the previous Commission's allocation.
- This corpus is split into two components:
 - State Disaster Response Fund (SDRF) — ₹1.6 lakh crore, meant for immediate response, relief, and reconstruction following a disaster
 - State Disaster Mitigation Fund (SDMF) — the remaining amount, meant to fund longer-term interventions that reduce the risk of a climate event escalating into a disaster
- These funds are available to all States, with inter-State distribution determined by a formula incorporating a disaster risk index (accounting for hazard frequency, intensity, exposure, and vulnerability) along with States' expenditure during the previous Commission's tenure.

The funds can be only used for notified disasters

Notified Disasters : The notified list has expanded from 12 to 14 categories, with heatwaves and lightning added to cyclone, drought, earthquake, fire, flood, tsunami, hailstorm, landslide, avalanche, cloudburst, pest attack, and frost and cold wave.

What if it is not a notified disasters : States can use upto 10% of their funds

The High Cost of India's Private Health-Care Boom

The high cost of India's private health-care boom

For crores of Indian families, the financial shock of hospitalisation – especially in a private hospital – begins even before treatment is complete. The Parliamentary Standing Committee on Health and Family Welfare, in its 176th Report tabled in Parliament on August 7, 2026, noted that the average cost of hospitalisation is ₹50,508 in a private facility, compared with ₹6,631 in a government facility. For childbirth, the average out-of-pocket medical expenditure in private facilities is ₹37,630, against ₹2,299 in public facilities.

Some of the contradictions

The committee has made 368 recommendations, including standardised package rates and mandatory pre-treatment cost estimates. Among the widely reported proposals is that basic room tariffs in metropolitan private hospitals should not exceed the average tariff of three-star hotels that are nearby. It has also suggested that large corporate hospitals earning from medical tourism, foreign patients and high-net-worth individuals should cross-subsidise poorer Indians and reserve beds for Ayushman Bharat-Pradhan Mantri Jan Arogya Yojana (AB-PMJAY) beneficiaries at regulated rates. Yet, there is a contradiction. India wants more private and foreign capital in health care, particularly in Tier-2, Tier-3 and rural areas, while simultaneously asking the government to review foreign direct investment rules related to the acquisition and management of existing hospitals.

There is a strong case for continued investment in Indian health care. Hospitals require substantial capital. Land, equipment, intensive care units, digital systems, laboratories and trained personnel are expensive. Public hospitals cannot currently meet all demand for secondary and tertiary care. Private hospitals therefore fill an important gap. Foreign investors and private-equity funds can bring capital, managerial capacity, technology and the capacity to expand hospital networks. India also needs investment in pharmaceutical and medical-device manufacturing. If regulation becomes excessively restrictive or unpredictable, some investment may move elsewhere. This could slow capacity expansion and delay the development of better health-care infrastructure in cities and districts where need is genuine.

However, in all sectors, capital comes with expectations of returns. In health care, it creates a special concern because patients are not ordinary consumers. A patient rarely decides independently whether an MRI is required, whether hospital admission should continue for two more days, or whether a procedure is necessary. Health care is shaped by what



Chandrakant Lahariya

Practising physician specialising in preventive and cardio-metabolic medicine, whose upcoming book, 'Pill-Free: You Don't Need Everything You Have Been Prescribed', will be released in September 2026

The country needs private capital to expand health care, but investment cannot be allowed to shape clinical decisions, pricing and access in ways that make care unaffordable or unnecessarily intensive

economists call information asymmetry: the provider knows more than the patient. Therefore, when financial incentives become too strong, they can influence not just the price of care but also how much care is delivered.

A closer look at private care

This is where the role of private equity and venture capital in health care deserves closer attention. Corporate hospital groups increasingly compete for well-known specialists, sophisticated technology and premium infrastructure. These can improve quality, but also create a high-cost ecosystem. Hospitals that pay very high salaries to senior specialists and super-specialists, acquire expensive equipment and operate under pressure to increase revenue must eventually recover these costs. Revenue targets, procedure-linked incentives, higher occupancy expectations and higher revenue per bed can gradually influence institutional behaviour. Most doctors act in patients' interests. Yet, systems shape behaviour. If hospitals and doctors are rewarded for doing more, the direction becomes predictable.

The result is not only higher cost but also a worrying medicalisation. Lab investigations may detect abnormalities that would never have caused harm. More screening can lead to unnecessary follow-up tests. A patient who could be managed as an outpatient may be admitted. Caesarean sections, angioplasties, intensive-care admissions, diagnostic packages and long lists of medicines must therefore be seen not only as individual clinical decisions but also within the incentive structure of the health system. This does not mean these interventions are usually unnecessary. It means that a system rewarding interventions and procedures needs strong safeguards against unnecessary prescriptions, surgeries, investigations and medications.

That is why a review of FDI in the acquisition of existing hospitals is justified. Domestic investors can be equally profit-oriented, while foreign investment can create valuable capacity. The more useful question is about what an investment does to the health system. Does it create new beds or simply acquire existing ones? Does it improve competition or lead to market concentration? Does it enter an underserved district or add another high-end facility in a metro? If an investor receives concessional land, tax benefits or other public support, are there enforceable obligations related to affordable beds or participation in public insurance schemes? India should encourage greenfield investment and manufacturing, while scrutinising acquisitions that reduce competition or raise the risk of excessive pricing.

Price regulation also needs to be approached carefully. The proposal to link hospital room charges to nearby three-star hotels is easy to

understand, but cannot by itself solve the affordability problem. A hospital room includes nursing, infection-control and emergency support that a hotel room does not. More importantly, if one component of the bill is capped, hospitals may increase charges elsewhere. India's experience with coronary stent price regulation showed that government intervention can reduce excessive mark-ups. But hospital care is more complex; what matters is the total cost of an episode. Package rates, transparent estimates, billing standards and audit mechanisms are therefore more useful than isolated caps on individual components. There is also an established approach to such payments: Diagnosis-Related Groups (DRG). A DRG is a patient-classification system that standardised hospital reimbursement by paying a fixed, predetermined amount for an inpatient stay based on the diagnosis and procedures, rather than reimbursing each service separately.

Focus on public health

There is a larger policy question. India cannot regulate its way out of weak public health care. If government hospitals remain overcrowded, understaffed or difficult to access, citizens will continue to depend heavily on private providers. Experience from OECD countries has shown that one effective form of regulation is a strong public health system that provides a credible alternative.

Public hospitals must therefore become a genuine option, not merely the last resort for those who cannot afford private care. Primary health care needs strengthening so that disease is prevented, detected and treated early. Insurance systems, including AB-PMJAY, should reward appropriate care rather than simply higher volumes of procedures. Clinical audits, evidence-based treatment protocols and transparent billing can protect both patients and doctors from commercial pressures.

The Parliamentary Committee is right to focus attention on affordability and to question whether the current pattern of investment in private health care aligns with public interest. Price caps and an FDI review are reasonable starting points, but India needs a wider conversation about the kind of health-care system it is building. Private investment will remain essential, and profit by itself is not the problem.

The problem begins when the pursuit of returns starts a key influence. A health-care system should attract capital, but its clinical priorities must still be set by medical need. The true measure of India's health-care progress will not be how much money flows into hospitals or what is the annual compounded growth rate in medical tourism, but whether Indian citizens can enter them with confidence that they will receive what they need. No more, no less.

The High Cost of India's Private Health-Care Boom

₹50,508 vs ₹6,631

**Average hospitalisation cost: private vs
government facility**

₹37,630 vs ₹2,299

**Average out-of-pocket childbirth cost: private
vs public facility**

Thank You